CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 2: Banking (Recurring Deposit Accounts)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the interest on a recurring deposit of ₹500 per month for 12 months at 8% per annum.

2. Find the maturity value of the account in the previous question.

3. Anil deposits ₹1,200 per month for 15 months at 6% per annum. Find the maturity value.

4. Meena deposits ₹800 per month for $1\frac{1}{2}$ years at 10% per annum. Find the amount she gets at maturity.

5. A recurring deposit account of ₹2,500 per month for 12 months earns ₹1,625 as interest. Find the rate of interest.

6. The maturity value of a recurring deposit account held for 3 years at 8% per annum is ₹12,132. Find the monthly deposit.

7. Sonia deposits ₹400 per month at 9% per annum and receives ₹198 as interest. Find the time in months.

8. A man deposits ₹1,000 per month in a recurring deposit account and gets ₹26,000 at the end of 2 years. Find the rate of interest.

9. Which gives more interest: ₹500 per month for 24 months at 6%, or ₹1,000 per month for 12 months at 6%?

10. Explain why the number of months in the interest formula appears as $\frac{n(n+1)}{2}$.

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Answer Key

1. $500 \times \frac{12 \times 13}{24} \times \frac{8}{100} = ₹260$
2. $6000 + 260 = ₹6{,}260$
3. $I = 1200 \times 10 \times 0.06 = ₹720$; M.V. $= ₹18{,}720$
4. $I = 800 \times \frac{18 \times 19}{24} \times 0.1 = ₹1{,}140$; M.V. $=$ ₹15,540
5. $1625 = 2500 \times 6.5 \times \frac{r}{100}$, so $r = 10\%$
6. $36P + 4.44P = 40.44P = 12132$, so $P = ₹300$
7. $198 = 400 \times \frac{n(n+1)}{24} \times 0.09$ gives $n(n+1) = 132$, so $n = 11$ months
8. Deposited ₹24,000; $I = ₹2{,}000 = 1000 \times 25 \times \frac{r}{100}$, so $r = 8\%$
9. $500 \times 25 \times 0.06 = ₹750$ and $1000 \times 6.5 \times 0.06 = ₹390$; the first
10. The deposits earn interest for $n, n-1, \ldots, 1$ months, and $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

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